Local Orbit Consistency for Symmetry-Robust Neural Operators

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Abstract

Neural operators can learn accurate PDE solution maps while still violating known symmetries of the underlying equations. We study stochastic local orbit consistency, a training objective that compares the prediction on a transformed input with the transformed original prediction and requires only model evaluations and known group actions. It requires no architectural change, supports labeled training and unlabeled adaptation, and leaves inference identical to the base operator. In a periodic N64 2D Navier–Stokes Galilean setting, augmentation plus orbit consistency reduces symmetry-induced OOD error and equivariance defect relative to supervised and augmentation baselines, with the strongest gains in low-label regimes. A five-seed reproduction with a near-capacity-matched DeepONet lowers OOD error by 7.0% and equivariance defect by 29.6% relative to its augmentation baseline, supporting portability to a second operator family while revealing backbone-dependent effect size.

1 Introduction

Neural operators are attractive for PDE learning because they approximate maps between function spaces rather than single finite-dimensional predictions. This flexibility also exposes a weakness: a high-accuracy operator need not respect symmetries that the exact solution operator satisfies. When test inputs move along a known symmetry orbit, such as a Galilean boost of a periodic flow, ordinary supervised training can produce both higher prediction error and larger equivariance defect.

The practical motivation extends beyond benchmark PDEs. Global atmospheric forecasting operates on spherical fields, where flat longitude–latitude operators can introduce coordinate artifacts and geometry-aware operators improve stability [3]. Neural operators are also used for vehicle-surface pressure and drag [16], frame-consistent tensorial turbulence closures [10], and plasma surrogates trained on simulated MHD and experimental tokamak observations [9]. At atomistic scale, forces must co-rotate with molecular coordinates, motivating $E(3)$-equivariant interatomic models [2]. These are motivations rather than applications established here: an action is admissible only when geometry, state, boundary and forcing data, diagnostic coordinates, and vector or tensor outputs transform jointly. Fixed geography, walls, or observation geometry may reduce or break a nominal symmetry.

We propose to reduce this failure mode with local orbit consistency. Given a known group action on inputs and outputs, the objective samples a small finite transform and penalizes the discrepancy between $\mathcal{G}_\theta(T_g a)$ and $T_g \mathcal{G}_\theta(a)$. The penalty is a training-time regularizer, not a new inference pipeline. This makes it complementary to supervised Lie augmentation and architecture-level equivariance: it can be added to an existing FNO, applied in low-label settings, and evaluated on unlabeled inputs.

Our empirical claim is deliberately narrow. The strongest evidence is the main N64 2D Galilean experiment, where augmentation plus orbit consistency with the same FNO2d backbone lowers orbit OOD relative L^2 and equivariance defect compared with forward-evaluation-matched supervised augmentation. A near-capacity-matched, fixed-grid DeepONet reproduction finds the same direction on that benchmark but a smaller predictive-error reduction. Burgers and D4 provide supporting evidence in other symmetry settings, and a fixed-molecule MLP force predictor provides a secondary mechanism check beyond Fourier layers; none establishes uniform gains across arbitrary equations or architectures. The core N64 artifact path, manifest, tables, and figures are regenerated by `scripts/reproduce_paper_artifacts.py`; reviewer-control tables use the separately documented `scripts/run_reviewer_followup_experiments.ps1` workflow.

2 Related Work

Neural operators learn mappings between function spaces and are therefore a natural model class for parametric PDE solution operators. DeepONet introduced a branch–trunk architecture for learning nonlinear operators from function samples and output coordinates [18]. Fourier Neural Operators (FNOs) parameterize global integral kernels in Fourier space and provide an efficient backbone for PDE families on regular grids [14]. The general neural-operator framework formalizes these models as discretization-invariant compositions of integral operators and nonlinearities, with applications to Burgers' equation, Darcy flow, and Navier–Stokes [12]. Physics-Informed Neural Operators (PINOs) add PDE residual constraints to operator learning, combining supervised operator regression with physics-based optimization [17]. Subsequent work extends neural operators to geometric settings: Geo-FNO learns deformations from irregular physical domains to latent uniform grids [15], GINO combines graph and Fourier operator components for large-scale three-dimensional PDEs on varying geometries [16], and NORM uses Laplacian eigenfunctions to define neural operators on Riemannian manifolds [5]. Operator-learning benchmarks such as PDEBench and The Well provide reusable datasets and baselines for time-dependent PDEs and broader physical simulation families [19, 22]. This literature establishes FNO-style models as computationally efficient solution-operator backbones, while also showing that domain geometry, discontinuities, and discretization effects require additional structure [13].

Symmetry has been used as an inductive bias through architectures, losses, and data transformations. Group-equivariant convolutional networks replace ordinary convolution by group convolution to share weights across symmetry orbits [7]; steerable CNNs generalize this principle through representation-constrained kernels [24]; and gauge-equivariant CNNs extend equivariance from global transformations to local frame changes on manifolds [8]. For operator learning, Group-Equivariant FNOs encode translation, rotation, and reflection symmetries in Fourier layers [11]. A parallel line derives symmetries from the governing differential equations. Lie Point Symmetry Data Augmentation constructs solution-preserving transformations from a PDE's Lie point symmetry group and uses them to reduce sample complexity for neural PDE solvers [4]. Lie-symmetry losses for PINNs enforce conservation of solution families under infinitesimal generators [1]. Generalized Lie symmetries for PINOs further argue that evolutionary representatives can provide stronger training signals than point symmetries alone [23]. These works support the central premise that PDE symmetries should constrain the learned solution operator, not only the training data distribution.

Model-agnostic symmetry methods provide the closest precedent for the proposed direction. Tangent propagation used transformation tangents to impose local invariance constraints on neural networks [21]. Lie Algebra Canonicalization uses infinitesimal generators to construct canonicalized inputs for pretrained models and neural operators under arbitrary Lie groups, including non-compact groups commonly arising in PDEs [20]. PACE-FNO applies a related canonicalization strategy to FNOs under symmetry-induced shifts, estimating a Lie-algebraic input frame, predicting in a canonical frame, and restoring the output frame for translations, rotations, and Galilean boosts [25]. LOCO occupies a distinct point in this design space. Unlike supervised Lie augmentation, its consistency term needs no transformed label; unlike hard-equivariant layers, it does not restrict the backbone; unlike canonicalization, it does not alter inference; and unlike tangent propagation, it uses ordinary evaluations at finite group elements rather than an explicit Jacobian–vector product at the identity. Relative to a plain finite-transform consistency loss, the local $\epsilon^2 + \eta$ normalization removes the leading quadratic dependence of the loss scale on step magnitude. The revision tests the finite-versus-tangent distinction and reports a scale-matched unnormalized control rather than treating these objectives as interchangeable.

3 Method

We train a neural operator $\mathcal{G}_\theta : \mathcal{A} \to \mathcal{U}$ with ordinary supervised regression and a stochastic local orbit-consistency objective. Each minibatch contains input–solution pairs (a, u). The supervised term is

$$\mathcal{L}_{\text{sup}}(\theta) = \mathbb{E}_{(a,u)} \left[\frac{\|\mathcal{G}_\theta(a) - u\|_2}{\max(\|u\|_2, 10^{-8})} \right]. \tag{1}$$

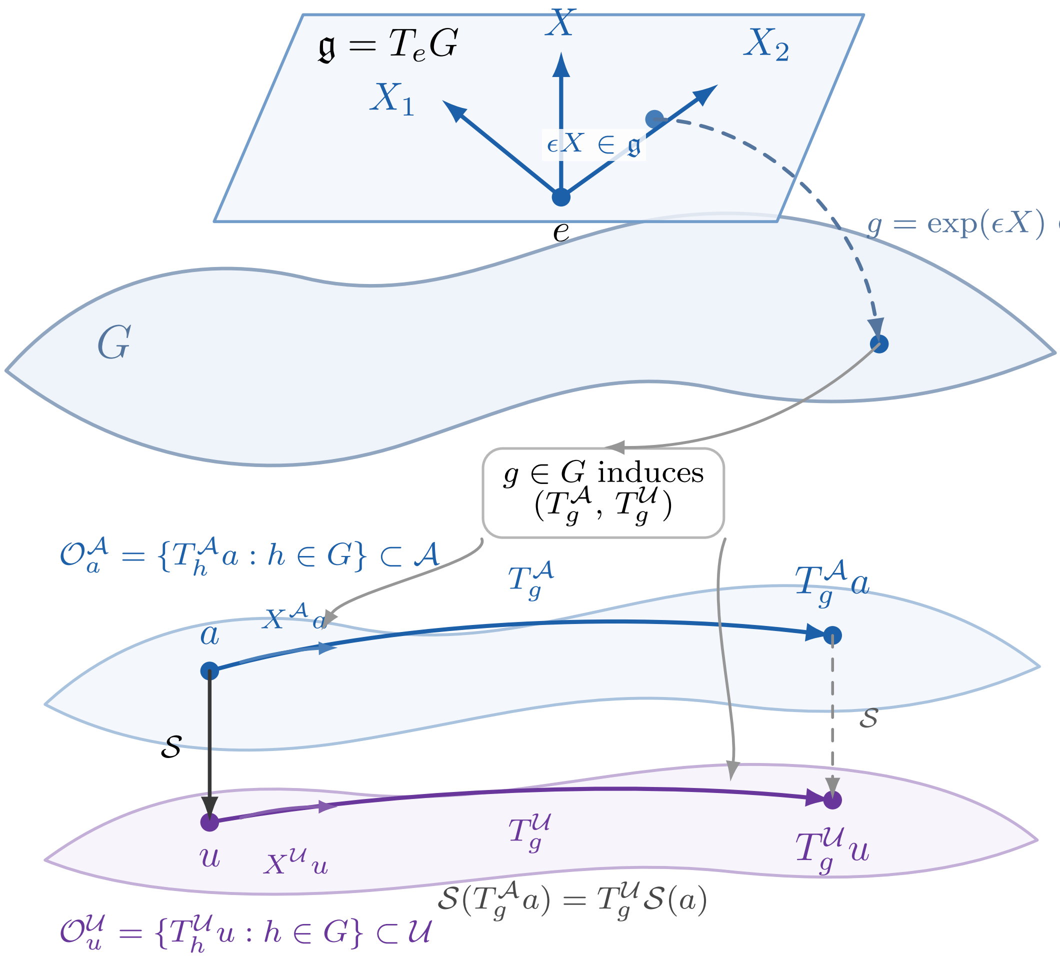

Figure 1: Lie-group actions on input and output field spaces. A local Lie-algebra step $\epsilon X \in \mathfrak{g} = T_eG$ maps through the exponential to a group element $g = \exp(\epsilon X) \in G$. The same g induces paired actions $T_g^{\mathcal{A}}$ and $T_g^{\mathcal{U}}$ on the input and output orbits. For an equivariant exact solution operator $\mathcal{S}$, the diagram commutes: $\mathcal{S}(T_g^{\mathcal{A}} a) = T_g^{\mathcal{U}} \mathcal{S}(a)$.

For a sampled Lie-algebra direction and finite transform g, let ϵ denote the step magnitude (for the two-dimensional Galilean action, $\epsilon = \max(\|\delta\|_2, 10^{-6})$). The implemented local orbit-consistency term is

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a,g} \left[\frac{\text{MSE}\big(\mathcal{G}_\theta(T_g^{\mathcal{A}} a) - T_g^{\mathcal{U}} \mathcal{G}_\theta(a)\big)}{\epsilon^2 + \eta} \right], \tag{2}$$

where MSE averages over all spatial and output entries of each example before the batch average. Supervised Lie augmentation uses the same unsquared per-example relative loss,

$$\mathcal{L}_{\text{aug}}(\theta) = \mathbb{E}_{(a,u),g} \left[\frac{\|\mathcal{G}_\theta(T_g^{\mathcal{A}} a) - T_g^{\mathcal{U}} u\|_2}{\max(\|T_g^{\mathcal{U}} u\|_2, 10^{-8})} \right]. \tag{3}$$

The combined method minimizes $\mathcal{L}_{\text{sup}} + \lambda_{\text{aug}} \mathcal{L}_{\text{aug}} + \lambda_{\text{orb}} \mathcal{L}_{\text{orb}}$; terms not used by a method are omitted.

What locality and step normalization add. A generic finite-transform consistency penalty would instead minimize $\mathbb{E}_{a,g}[\text{MSE}(\Delta_\theta(a, g))]$, without dividing by the local step. Near the identity, $\Delta_\theta(a, \exp(\epsilon X)) = \epsilon d_\theta(a; X) + O(\epsilon^2)$. The unnormalized penalty therefore scales as $\mathbb{E}[\epsilon^2 \|d_\theta\|^2] + O(\mathbb{E}|\epsilon|^3)$: larger sampled steps

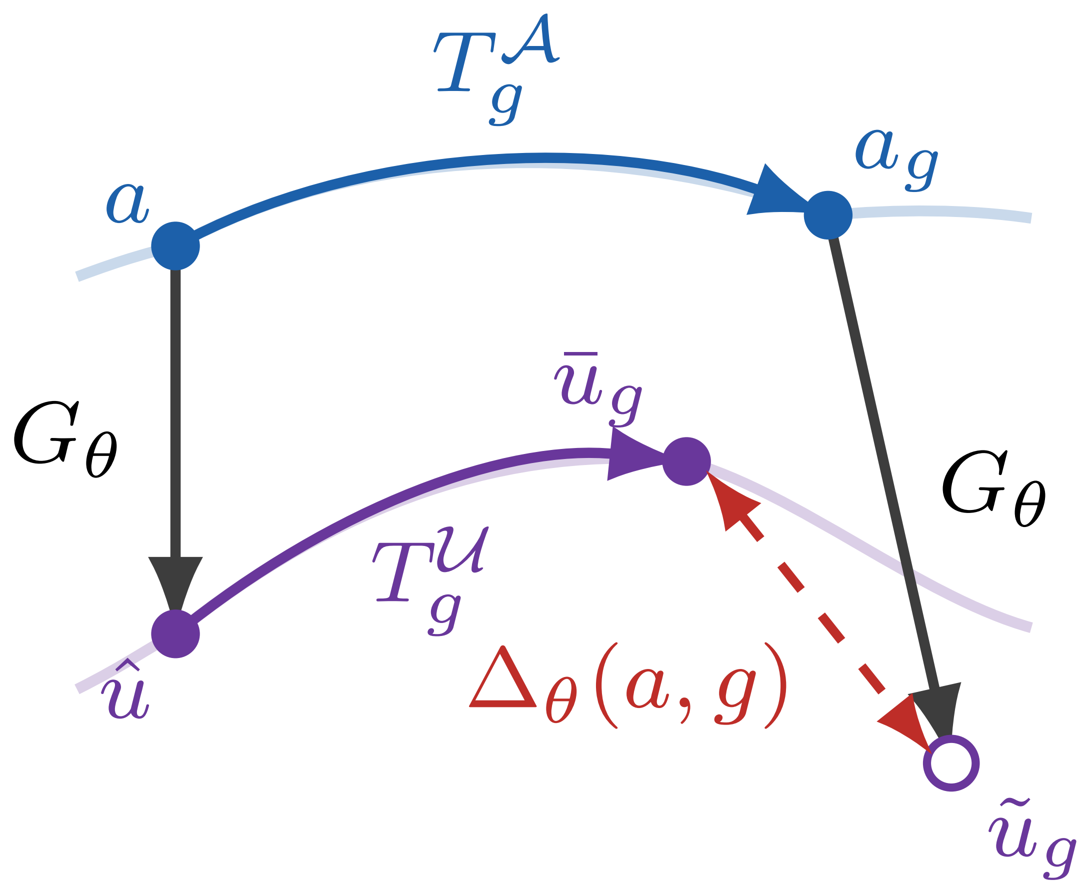

Figure 2: Local orbit consistency. For $g = \exp(\epsilon X)$, let $a_g = T_g^{\mathcal{A}} a$, $\hat{u} = G_\theta(a)$, $\bar{u}_g = T_g^{\mathcal{U}} \hat{u}$, and $\tilde{u}_g = G_\theta(a_g)$. LOCO penalizes the red equivariance defect $\Delta_\theta(a, g) = \tilde{u}_g - \bar{u}_g$, equivalently $\mathrm{MSE}(\tilde{u}_g - \bar{u}_g)/(\epsilon^2 + \eta)$ in the discrete implementation.

receive quadratically greater leading-order weight, and the loss scale changes when the transform radius changes. Dividing by $\epsilon^2 + \eta$ removes this leading scale when $\epsilon^2 \gg \eta$, so the weight is interpretable against a tangent defect and is less mechanically coupled to the chosen local radius. The stabilizer caps the weight near zero. This normalization does not discard finite-orbit information: at nonzero steps the exact nonlinear group action retains higher-order terms that tangent propagation at the identity does not see. Section 6 compares the normalized objective with both explicit tangent propagation and a scale-matched unnormalized finite-transform control.

The loss is architecture-agnostic in the implementation sense: it requires only model evaluations and known input/output actions, not Fourier layers. We test it primarily with FNO backbones, reproduce the main Galilean comparison with a near-parameter-matched DeepONet, and separately use a non-equivariant molecular MLP as a mechanism check. This does not imply backbone-independent effect size. Transformations are applied to physical-grid input fields; the prediction on the transformed input is compared with the transformed original prediction; gradients pass through both model evaluations. Integer periodic translations are tensor rolls, non-integer periodic translations are spectral shifts, and finite square symmetries are exact tensor rotations and flips. For 1D Burgers, the Galilean action on the solution map $u_0 \mapsto u_T$ is $T_c^{\mathcal{A}} u_0(x) = u_0(x) + c$ and $T_c^{\mathcal{U}} u_T(x) = u_T(x - cT) + c$.

The same loss supports unlabeled adaptation. Given a pretrained operator $\mathcal{G}_{\theta_0}$ and target inputs $a \sim \mathcal{D}_{\mathrm{target}}$, we minimize $\mathcal{L}_{\mathrm{orb}}^{\mathrm{target}}(\theta) + \beta \|\theta - \theta_0\|_2^2$ over a chosen adaptation set Θ_{adapt}, such as all parameters, the last operator block, or the projection head. This adaptation objective requires no PDE solves and no target labels.

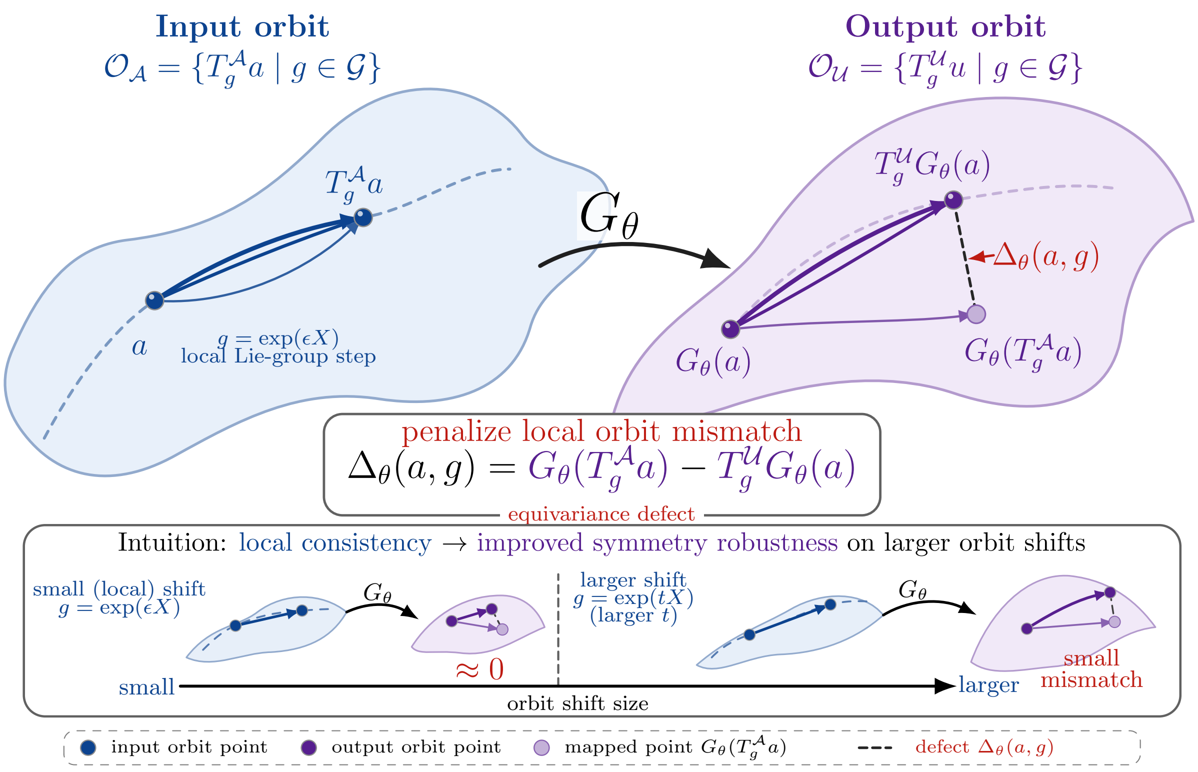

Figure 3: Vector schematic of local orbit consistency. A local Lie-group step moves an input a to $T_g^{\mathcal{A}} a$. The training penalty compares the prediction on the transformed input, $\mathcal{G}_\theta(T_g^{\mathcal{A}} a)$, with the transformed original prediction, $T_g^{\mathcal{U}} \mathcal{G}_\theta(a)$, reducing the equivariance defect along the orbit while leaving inference unchanged.

4 Theory

Let $\mathcal{A}$ and $\mathcal{U}$ denote Banach spaces of input and output fields. A PDE family induces a solution operator $\mathcal{S} : \mathcal{A} \to \mathcal{U}$. Let G be a Lie group acting on the full problem family through transformations $T_g^{\mathcal{A}}$ and $T_g^{\mathcal{U}}$. Closure of the full problem class under G means that the equation, coefficients, forcing terms, boundary or initial conditions, domain class, and observation map transform consistently. Under this assumption the exact solution operator is equivariant: $\mathcal{S}(T_g^{\mathcal{A}} a) = T_g^{\mathcal{U}} \mathcal{S}(a)$. For a learned operator $\mathcal{G}_\theta$, define the operator-level equivariance defect $\Delta_\theta(a, g) = \mathcal{G}_\theta(T_g^{\mathcal{A}} a) - T_g^{\mathcal{U}} \mathcal{G}_\theta(a)$. Our regularizer jointly samples a Lie-algebra direction X, a step ϵ, and $g = \exp(\epsilon X)$, then penalizes

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a,X,\epsilon} \left[\frac{\|\Delta_\theta(a, \exp(\epsilon X))\|_{\mathcal{U}}^2}{\epsilon^2 + \eta} \right]. \tag{4}$$

For gridded experiments, $\|\cdot\|_{\mathcal{U}}^2$ is discretized as the mean square over spatial and output entries, matching the implementation in Section 3. The objective is label-free because both terms are model predictions transformed by known group actions. For the N64 square sampling law, $X = \delta / \|\delta\|_2$ away from the measure-zero identity and $\epsilon = \|\delta\|_2$; direction and step are therefore jointly induced rather than independently sampled.

Infinitesimal interpretation. Assume $T_{\exp(\epsilon X)}^{\mathcal{A}}$, $T_{\exp(\epsilon X)}^{\mathcal{U}}$, and $\mathcal{G}_\theta$ are differentiable at $\epsilon = 0$. Let $X^{\mathcal{A}} a$ and $X^{\mathcal{U}} u$ denote the induced infinitesimal actions. Taylor expansion gives

$$\mathcal{G}_\theta(T_{\exp(\epsilon X)}^{\mathcal{A}} a) - T_{\exp(\epsilon X)}^{\mathcal{U}} \mathcal{G}_\theta(a) = \epsilon \left(D\mathcal{G}_\theta(a)[X^{\mathcal{A}} a] - X^{\mathcal{U}} \mathcal{G}_\theta(a) \right) + O(\epsilon^2). \tag{5}$$

Thus $\|\Delta_\theta(a, \exp(\epsilon X))\|_{\mathcal{U}}^2 / \epsilon^2$ converges to $\|D\mathcal{G}_\theta(a)[X^{\mathcal{A}} a] - X^{\mathcal{U}} \mathcal{G}_\theta(a)\|_{\mathcal{U}}^2$ as $\epsilon \to 0$. For the implemented denominator $\epsilon^2 + \eta$, this tangent interpretation applies when $\epsilon^2 \gg \eta$, or along a limit with $\eta = o(\epsilon^2)$; with fixed positive η, the loss instead vanishes as $\epsilon \to 0$. LOCO is therefore a finite-difference-based regularizer that avoids explicit Jacobian–vector products, not an unbiased tangent estimator at the finite radii used experimentally. For nonzero ϵ, the normalized defect also contains higher-order variation of the learned map and group orbit. Enforcing commutation over a sampled neighborhood can therefore constrain finite displacements that an identity-only directional derivative does not; the tangent and finite-step controls in Section 6 test this distinction.

Orbit error decomposition. If the true solution operator is equivariant and $T_g^{\mathcal{U}}$ is bounded, then for any $a \in \mathcal{A}$ and $g \in G$,

$$\begin{aligned}\|\mathcal{G}_\theta(T_g^{\mathcal{A}}a) - \mathcal{S}(T_g^{\mathcal{A}}a)\|_{\mathcal{U}} \le{}& \|\mathcal{G}_\theta(T_g^{\mathcal{A}}a) - T_g^{\mathcal{U}}\mathcal{G}_\theta(a)\|_{\mathcal{U}} \\ &+ \|T_g^{\mathcal{U}}\|\,\|\mathcal{G}_\theta(a) - \mathcal{S}(a)\|_{\mathcal{U}}.\end{aligned} \tag{6}$$

The first term is the learned equivariance defect and the second term is the in-distribution prediction error transported by the output action. This inequality motivates reducing $\mathcal{L}_{\mathrm{orb}}$ when test inputs differ from training inputs by known PDE symmetries.

Approximate symmetries and cost. For partial or approximate symmetries, replace the output norm by a weighted norm $\|v\|_w^2 = \|w_{a,g} \odot v\|^2$, where $w_{a,g}$ can encode valid overlap after transformations, boundary exclusions, observation masks, or regions where transformed coefficients and forcing terms remain physically meaningful. With one sampled transform per minibatch, the LOCO term uses one evaluation on the original input, one on the transformed input, and a known output transformation. At deployment there is no LOCO module or additional operator call: inference uses the same base graph $\mathcal{G}_\theta(a)$, and the measured latency differences are within run-to-run variation.

5 Experimental Details and Evaluation Protocol

This section gives the complete protocol behind the reported results. We distinguish three quantities that were conflated in the shorter submission: the ambient frame used to generate each Navier–Stokes example, the relative transform sampled during training, and the relative transform used for the symmetry-shifted test evaluation. All "radii" below are componentwise bounds. Thus $\delta \sim \mathcal{U}([-r, r]^2)$, not a uniform draw from a Euclidean ball, and the step used in the LOCO denominator is $\epsilon = \max(\|\delta\|_2, 10^{-6}) \le \sqrt{2}r$. The clamp is a numerical guard for the tiny region $\|\delta\|_2 < 10^{-6}$, not only the exact identity.

5.1 Headline N64 Galilean experiment

Equation, domain, and numerical solver. We solve the unforced vorticity equation

$$\partial_t\omega + v \cdot \nabla\omega = \nu\Delta\omega, \qquad v = (\partial_y\psi, -\partial_x\psi) + c, \qquad \Delta\psi = \omega, \tag{7}$$

on the periodic square $[0, 1)^2$. This sign convention is the one used by the released solver. The solution is represented on a 64×64 uniform grid and advanced from $t = 0$ to $T = 0.5$ with a pseudo-spectral classical fourth-order Runge–Kutta method, time step 10^{-3} (500 steps), viscosity $\nu = 10^{-3}$, tensor-product 2/3-rule de-aliasing of the nonlinear term, and no forcing. Solver generation uses batches of eight only to limit memory; it does not change the numerical method.

Initial conditions and ambient frames. For every example, we draw independent standard Gaussian noise z on the grid, apply the Fourier filter $\widehat{\omega}_0(k) = \widehat{z}(k)\exp(-\|k\|_2^2/8)$, subtract the spatial mean, divide by the sample spatial standard deviation plus 10^{-6}, and use amplitude one. We then draw the constant ambient-velocity components independently as $c_x, c_y \sim \mathcal{U}[-0.5, 0.5]$. The input channels are $(\omega_0, c_x, c_y) \in \mathbb{R}^{64\times64\times3}$, with the two scalar velocities broadcast over the grid; the target is the terminal vorticity $\omega(T) \in \mathbb{R}^{64\times64\times1}$. All stored tensors are `float32`. We apply no model-side input or target normalization, discard no examples, and use no additional data cleaning or clipping.

Table 1: Complete data-generation parameters for the headline boosted N64 Navier–Stokes dataset. Generation seed offsets separate the three independently generated splits.

Item	Value
Domain / grid	periodic $[0,1)^2$, 64×64
Train / validation / test	1,024 / 128 / 256
Dataset seed	31; split offsets 0 / 100,000 / 200,000
Final time / time step	$T = 0.5$ / $\Delta t = 10^{-3}$
Viscosity / forcing	10^{-3} / none
Integrator	pseudo-spectral RK4, 500 steps
Aliasing control	tensor-product 2/3 filter
Initial-condition filter	$\exp(-\|k\|_2^2/8)$
IC centering / scaling	zero spatial mean; SD $+10^{-6}$; amplitude 1
Ambient velocity	$c_x, c_y \overset{\text{iid}}{\sim} \mathcal{U}[-0.5, 0.5]$
Generator batch / precision	8 / `float32`
Input / target shapes	$64 \times 64 \times 3$ / $64 \times 64 \times 1$
Processed SHA-256	`defc9703dbc41494fabf9fcfe9e3408eb2e1c38908` `14e47fe03d0f6b849e3f50`
Config fingerprint	`b2b5dec2dd3049ff491071674c34042c89f8c1a855` `f86cbb199614bfafed0cdd`

Splits and labeled subsets. The train, validation, and test fields are generated independently and stored before model fitting. Validation and test splits are always used in full. For a training fraction $q < 1$, a run with seed s retains the first $\max(1, \mathrm{round}(1024q))$ indices of a `randperm` generated with seed s. The resulting 1%, 2%, 5%, and 10% subsets therefore contain 10, 20, 51, and 102 labeled examples. Methods with the same seed and fraction use the same labeled subset. Consequently, variation across training seeds includes both parameter initialization and labeled-subset variation, whereas the generated dataset itself remains fixed.

FNO2d architecture. Every headline row uses the same FNO2d with three input channels and one output channel, width 48, 12 retained modes on each axis, and four Fourier blocks. Each block sums a learned spectral convolution and a 1×1 pointwise convolution and applies GELU. A pointwise $48 \to 96 \to 1$ projection with GELU produces the output. We use no coordinate channels, spatial padding, normalization layers, or dropout. The model has 2,668,609 trainable parameters. LOCO adds no parameters or inference-time operation.

DeepONet architecture and backbone-control protocol. The Reviewer 2 backbone control uses a fixed-grid DeepONet with the same 64×64, three-channel inputs and one-channel outputs. Its branch encoder has four circularly padded 3×3, stride-2 convolutions with channel widths $3 \to 32 \to 64 \to 128 \to 256$, GELU after every convolution, and a $4096 \to 480 \to 256$ GELU projection. We deliberately use no normalization: normalizing spatially constant channels could erase the two ambient-velocity inputs. The trunk receives 48 fixed periodic coordinate features, $\{\sin(2\pi kx), \cos(2\pi kx), \sin(2\pi ky), \cos(2\pi ky)\}_{k=1}^{12}$, and applies a $48 \to 256 \to 256 \to 256 \to 256$ MLP with GELU after the first three linear maps. A rank-256 branch–trunk inner product, divided by $\sqrt{256}$, plus one learned scalar bias produces each output-grid value. The model has 2,688,033 trainable parameters, 0.73% more than the FNO2d. The circular padding and periodic query features encode the domain topology; there is no spectral convolution and no hard-coded Galilean input–output action. The sensor and query grids are fixed at N64, so this experiment does not test resolution transfer.

We reproduce all four headline training methods with seeds $\{23, 31, 47, 59, 71\}$ and predeclare DeepONet augmentation versus DeepONet augmentation plus normalized LOCO as the primary paired comparison. It shares the dataset, seed-indexed 20-example labeled subsets, transform distribution, optimization hyperparameters, and validation rule with the FNO protocol. All four DeepONet methods use the same DeepONet inference graph and parameter count. Methods share the initialization seed and labeled subset, but their full stochastic training trajectories are not common-random-number coupled: the schedules restart the 20-example loader a different number of times, and the LOCO method consumes an additional independent

Table 2: Headline N64 training variants. At 2% labels, batches cycle through sizes 16 and 4. The transformed-example totals below use the actual partial batches rather than treating every update as a full batch.

Method	Steps/epoch	Updates	Batch forwards	λ_{orb}	Aug. examples	Orbit examples
FNO	4	600	600	–	0	0
FNO+LOCO	4	600	1,200	0.05	0	6,000
FNO+augmentation	6	900	1,800	–	9,000	0
FNO+augmentation+LOCO	4	600	1,800	0.10	6,000	6,000
DeepONet	4	600	600	–	0	0
DeepONet+LOCO	4	600	1,200	0.05	0	6,000
DeepONet+augmentation	6	900	1,800	–	9,000	0
DeepONet+augmentation+LOCO	4	600	1,800	0.10	6,000	6,000

transform draw per update. Validation and test transform draws are seed-matched across methods. We fixed the architecture and copied the FNO coefficients 0.05 (LOCO only) and 0.10 (augmentation plus LOCO) before completing the five-seed test outcomes; there was no DeepONet-specific test-set or lambda sweep.

Optimization and checkpointing. Training uses `float32` without automatic mixed precision. We optimize with AdamW for 150 epochs using learning rate 10^{-3}, weight decay 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$, optimizer $\epsilon = 10^{-8}$, and no AMSGrad. A cosine-annealing schedule spans all 150 epochs, has zero minimum learning rate, and uses no warm-up. We clip the global gradient norm at 1.0. The shuffled training loader uses batch size 16, no workers, and retains the final partial batch; validation and test loaders are not shuffled. Validation runs every 10 epochs on all 128 examples. We do not early-stop and select the checkpoint with the lowest validation ID mean per-example relative L^2.

The supervised and augmented terms are the implemented unsquared per-example relative L^2 losses in Section 3, with $\lambda_{\text{aug}} = 1$. The finite-orbit numerator is a mean square over spatial/output entries and is divided by $\epsilon^2 + \eta$, with $\eta = 10^{-6}$. On every relevant update, augmentation and LOCO draw independent transforms; within each draw, transform parameters are sampled independently for every example. Gradients propagate through both LOCO prediction branches.

The primary fixed-weight comparison uses $\lambda_{\text{orb}} = 0.10$; the orbit-only row uses 0.05. The robustness sweep evaluates $\lambda_{\text{orb}} \in \{0.01, 0.05, 0.10, 0.20, 0.30\}$, and we identify 0.30 explicitly when reporting the post-hoc tuned row. Unless a table says otherwise, five-seed results use seeds $\{23, 31, 47, 59, 71\}$, while compact controls use $\{23, 31, 47\}$.

Galilean training and transformed-test distributions. For a relative boost $\delta = (\delta_x, \delta_y)$, the input action adds δ to the two ambient-velocity channels and the output action applies the exact periodic spectral shift $\omega_T(x) \mapsto \omega_T(x - \delta T)$. During ordinary training and test evaluation, $\delta_x, \delta_y \overset{\text{iid}}{\sim} \mathcal{U}[-0.25, 0.25]$. The severity study instead uses component bounds $r \in \{0.10, 0.20, 0.25, 0.35, 0.50\}$. These are additional relative boosts of the same held-out examples, not a separately simulated test split; because the original ambient frame is in $[-0.5, 0.5]^2$, the transformed observable frame at $r = 0.50$ can extend to $[-1, 1]^2$, although not every draw lies outside the original componentwise support. We therefore use "symmetry-shifted OOD" in this specific sense.

Evaluation, uncertainty, and latency. For each checkpoint, ID relative L^2 averages one per-example ratio over all 256 test examples. Orbit OOD error and relative equivariance defect average over four independent transforms per example (1,024 transformed pairs); the standard test transform RNG seed is $s+200{,}000$. The severity sweep also uses four draws per example, with seed $s+700{,}000$. Tables report the arithmetic mean and sample standard deviation (`ddof=1`) across seeds. Paired comparisons use matched-seed differences and two-sided 95% Student-t intervals with $n - 1$ degrees of freedom.

Latency is measured on the first 16-example test batch using five unmeasured warm-up forwards followed by 20 timed forwards; milliseconds per sample is elapsed time divided by 20×16. The retained headline manifests record CPU execution, Windows platform string `Windows-10-10.0.26100-SP0`, Python 3.11.6, and PyTorch 2.12.0+cpu. The host currently identifies its processor as an 11th Gen Intel Core i7-1185G7 at 3.00

Table 3: Secondary dataset generation, curation, and preprocessing. All processed tensors are `float32`.

Dataset	Domain, resolution, splits	Evolution parameters	Initial data or source	Preparation and processed SHA-256
1D viscous Burgers	Periodic $[0, 1)$, $N = 128$; 2,048/256/512	$T = 0.5$, $\Delta t = 10^{-3}$, $\nu = 0.01$; pseudo-spectral RK4, 500 steps, 2/3 de-aliasing; generator batch 64	Modes $k = 1, \ldots, 8$, independent Gaussian sine/cosine coefficients scaled by $k^{-1.5}$; zero mean, unit SD, amplitude 0.8, mean 0; dataset seed 23 and split offsets 0/100,000/200,000	No further normalization or exclusions; shapes $128 \times 1 \rightarrow 128 \times 1$; `1753fa2e5ca4fb474f93fc8cdef5732d aea94b7d36d7537113b38f9e3d029758`
Unboosted N64 Navier–Stokes (D4)	Periodic $[0, 1)^2$, 64^2; 1,024/128/256	Same T, Δt, ν, solver, de-aliasing, and generator batch as the headline data; no ambient velocity	Same filtered Gaussian vorticity law (smoothness 8, amplitude 1); dataset seed 31 and split offsets 0/100,000/200,000	One input and one target vorticity channel; no further normalization; `defade883926ad6ba6ef469bf1e6ba69 b3f7d57e743e4f53c83a05acb88a8f7e`
1D Dirichlet heat diagnostic	$[0, 1]$, $N = 64$ including endpoints; 512/128/128	$u_t = 0.01u_{xx}$, $T = 0.1$; exact eight-mode sine-series decay $e^{-0.01(\pi k)^2 T}$	Eight sine modes with independent Gaussian coefficients scaled by $1/k$, samplewise unit-SD normalization, amplitude 1; split seeds 23/34/45	Homogeneous Dirichlet endpoints; no further normalization; shapes $64 \times 1 \rightarrow 64 \times 1$; `7427ec21304ae8879e700653e65ee239 3416b2daa8bc4b27756c1cae1ff09b2d`
rMD17 ethanol [6]	Fixed nine-atom molecule; 500/1,000/1,000 and 1,000/1,000/1,000	Static force regression; no PDE solve	Released `rmd17_ethanol.npz`; raw SHA-256 `0824d78c687b8ede25f7162c54c5c643 4cfe37f9f859ee8f223c60379d7d5cf9`; historical custom selection with metadata seed 2027, rather than the official split CSVs	Subtract each conformation's arithmetic coordinate centroid; append nuclear charge $Z/10$; targets are unnormalized forces. Exact cached 500-label tensor is preserved in the artifact with raw `old_indices`; SHA-256 `fb5515b3355651459c2ad9078e7cfb5 bb63dce96e55e7c36956cbe9ff25e5472`. Coordinates in Å, forces in kcal mol^{-1}Å^{-1}

GHz; the original manifests did not record the CPU model, RAM, BLAS thread count, or peak memory, so latency should be interpreted as host-specific. Deterministic kernels were not forced (`deterministic=false`), but every run stores its resolved config, config hash, Git commit, dataset SHA-256, checkpoint, metrics, and initial/final Torch RNG states.

The DeepONet campaign uses the same host and software versions and records `OMP_NUM_THREADS=1` and `MKL_NUM_THREADS=1`. Its seed shards were executed concurrently as uninterrupted direct runs; checkpoint-chunk resume was not used. The recorded run wall times and latency samples are audit fields rather than controlled cross-method or cross-backbone efficiency comparisons. The unchanged-inference statement follows directly from using the identical DeepONet graph and parameter count with and without LOCO.

5.2 Secondary datasets and model hyperparameters

Table 3 records the complete preparation of the secondary benchmarks. These experiments are reported as scope and mechanism checks; the headline claim remains the boosted N64 setting above.

Burgers transform mixture. The reported Burgers orbit metrics are not Galilean-only. For each mini-batch, we choose either a periodic translation (probability 0.5, shift $s \sim \mathcal{U}[-0.25, 0.25]$) or a Galilean boost (probability 0.5, $b \sim \mathcal{U}[-0.35, 0.35]$); parameters within the selected family are then independent across examples. The Galilean action is $u_0(x) \mapsto u_0(x) + b$ and $u_T(x) \mapsto u_T(x - bT) + b$. Severity multipliers $1, 1.5, 2, 3$ produce translation bounds $0.25, 0.375, 0.50, 0.75$ and boost bounds $0.35, 0.525, 0.70, 1.05$. Standard evaluations use four draws per example; the severity analysis uses eight.

Molecular rigid motions. For rMD17, each transform draws an axis by normalizing a three-dimensional standard Gaussian vector, an angle uniformly in $[-\pi/2, \pi/2]$, and each translation component uniformly in $[-1, 1]$ Å. Coordinates rotate and translate, whereas force vectors rotate only. For this product action we disclose the implemented composite step $\epsilon = \max(|\theta| + \|t\|_2, 10^{-6})$, with angle in radians and translation in Å. This is a disclosed composite step weighting that mixes the configured angular and translational units; it is not a canonical, unit-free SE(3) norm. The reported LOCO row divides by $\epsilon^2 + 10^{-3}$ and uses $\lambda_{\text{orb}} = 0.03$, a rounded approximate average-scale match to the earlier raw coefficient 0.01 under this transform law. We use 10 latency warm-ups and 50 repeats. For the reported 500-label comparison, augmentation performs 24 updates per epoch with two model evaluations per update, whereas Aug.+LOCO performs 16 updates with three evaluations per update. Both therefore use 7,200 batch-level model evaluations over 150 epochs, although their optimizer and backward-pass counts differ. The historical cache predates the current rMD17

Table 4: Model and optimization hyperparameters for every reported benchmark. Settings not repeated in the last column use the AdamW/cosine defaults stated in the main protocol.

Benchmark	Architecture	Parameters	Epochs / batch	Benchmark-specific training and transforms
Boosted N64	FNO2d, $3 \to 1$, width 48, modes 12×12, depth 4, no grid, GELU	2,668,609	150 / 16	2% headline has 20 labels and 4 steps/epoch; augmentation compute match uses 6; $\lambda_{\mathrm{aug}} = 1$, fixed $\lambda_{\mathrm{orb}} = 0.10$, $\eta = 10^{-6}$
Boosted N64 / DeepONet	Circular CNN branch $3 \to 32 \to 64 \to 128 \to 256$, $4096 \to 480 \to 256$ head; 12-harmonic periodic coordinate trunk, width/rank 256	2,688,033	150 / 16	Same 20-label subsets, seeds, optimizer, transform law, and 4/4/6/4 step schedules as the FNO four-method protocol; fixed $\lambda_{\mathrm{orb}} = 0.10$ for the primary Aug.+LOCO row
Burgers	FNO1d, $1 \to 1$, width 64, 16 modes, depth 4, no grid, GELU	287,361	150 / 32	Full 2,048 labels, 64 steps/epoch; seeds 23/31/47; orbit-only $\lambda = 0.05$, Aug.+LOCO $\lambda = 0.25$
Dirichlet heat	FNO1d, $1 \to 1$, width 32, 12 modes, depth 4, coordinate grid, GELU	55,649	80 / 32	10%=51 labels, 2 steps/epoch; seeds 23/31/47; $\lambda_{\mathrm{aug}} = 1$, $\lambda_{\mathrm{orb}} = 0.05$, $\eta = 10^{-6}$
rMD17 ethanol	Flatten $9{\times}4$ input; four width-128 SiLU linear layers; 27-output linear head reshaped to 9×3	57,755	150 / 32	Reported 500-label comparison: Aug. 24 and Aug.+LOCO 16 steps/epoch (48 batch forwards/epoch), seeds 23/31/47/59/71; validation every 25 epochs; $\lambda_{\mathrm{aug}} = 1$, $\lambda_{\mathrm{orb}} = 0.03$, $\eta = 10^{-3}$; step normalization enabled
D4 FNO	FNO2d, $1 \to 1$, width 48, modes 12×12, depth 4, no grid, GELU	2,668,513	150 / 16	1%=10 labels, 4 steps/epoch; seeds 23/31/47/59/71; Aug.+LOCO $\lambda = 0.20$, $\eta = 10^{-6}$
D4 G-FNO	D4 group FNO, pseudoscalar input/output, width 8, modes 6×6, depth 4, no grid	350,280	150 / 16	Supervised only; same 10-label subsets, 4 steps/epoch, and five seeds; architectural comparator is not parameter- or mode-matched

generator implementation. Although its metadata records split seed 2027, a fresh `torch.randperm` does not reconstruct the same selection. The exact 500/1,000/1,000 tensor used here is therefore included as the authoritative evidence input, and its retained raw `old_indices` make the selected conformations auditable against the hashed NPZ. The released verification script matches all 2,500 cached conformations to the raw NPZ with zero `float32` input and force discrepancy. The older 1,000-label cache has different validation/test slices and is not used as a controlled label-scaling comparison.

Finite D4 action. For the D4 experiment, a rotation index is uniform on $\{0, 1, 2, 3\}$ and reflection is Bernoulli(0.5), giving all eight elements uniformly. Vorticity is treated as a pseudoscalar and changes sign under reflection. We set $\epsilon = 1$, so this is finite-group consistency rather than a local Lie-algebra step. Augmentation and Aug.+LOCO both use four updates per epoch; their forward-pass compute is therefore not matched.

Boundary diagnostic. The heat-equation shift is sampled as $s \sim \mathcal{U}[-0.1, 0.1]$ and implemented by zero-padded linear interpolation. The masked variant evaluates only source coordinates in $[0.05, 0.95]$. Because a fixed Dirichlet problem is not globally translation-equivariant and boundary influence propagates into the interior, this experiment is an approximate boundary-mask diagnostic, not an exact-symmetry OOD benchmark. Its transformed target is the shifted cached output, not a newly solved heat equation for the shifted input. To avoid comparing checkpoints on different scoring supports, we re-evaluate all three training variants with the same transform draws under both a full-domain score and a common valid-overlap score with margin 0.05. Masking restricts where this approximate constraint is scored; it does not restore translation equivariance of the fixed-boundary heat equation.

5.3 Additional N64 controls

Step-size and tangent controls. At 2% labels, the denominator sensitivity uses $\eta \in \{10^{-8}, 10^{-6}, 10^{-4}\}$ at component radius 0.25, while the step-size sensitivity uses radii $\{0.125, 0.25, 0.50\}$ at $\eta = 10^{-6}$; each uses seeds 23, 31, and 47. The tangent-propagation comparison uses the same 20 labels, four steps per epoch, all five headline seeds, and $\lambda_{\mathrm{tangent}} = 0.10$. Its input tangent changes only the two ambient velocity channels, and its output tangent is $-T(\delta_x \partial_x + \delta_y \partial_y)\mathcal{G}(a)$. A differentiable JVP is computed with graph construction enabled, and the tangent mean square is divided by $\|\delta\|_2^2+10^{-6}$. Augmentation+tangent draws its supervised augmentation independently.

The plain finite-transform control uses the same data, architecture, transform range, augmentation, and four-update schedule as normalized LOCO, but minimizes the raw finite equivariance MSE. Since $\delta_x, \delta_y \overset{\text{iid}}{\sim} \mathcal{U}[-0.25, 0.25]$, $\mathbb{E}\|\delta\|_2^2 = 2(0.25)^2/3 = 1/24$. We therefore set its raw-loss weight to 2.4, an approximate leading-order average-scale match to the normalized $\lambda_{\text{orb}} = 0.10$ objective. The square sampling law couples direction and norm, so this is not an exact equivalence, nor an assertion that the two nonlinear losses have identical realized magnitudes.

Semi-supervised joint training. The joint 2%-label experiment uses supervised and augmented batches from the 20-example labeled subset and separate LOCO batches from the full 1,024-example training pool. It performs four combined updates per epoch, uses full-pool orbit batch size 16, and therefore samples 64 (not all 1,024) full-pool orbit examples per epoch; $\lambda_{\text{orb}} = 0.05$ and seeds are 23, 31, and 47.

Unlabeled target adaptation. The target-adaptation table is transductive: the same 256 test inputs are used without labels during adaptation and with labels for before/after evaluation. Starting from the 2%-label augmentation or fixed-weight Aug.+LOCO checkpoint, we update only the FNO projection layers (4,801 parameters) for five epochs and 80 updates. We use AdamW with learning rate 10^{-5}, zero weight decay, no scheduler, batch size 16, and gradient clip 0.1. The objective combines orbit loss with a unit-weight relative prediction-preservation term; parameter anchoring has weight zero. Target and orbit boosts are independently redrawn at component radius 0.35 or 0.50. Evaluation uses four transforms per example and seeds 23, 31, and 47.

Canonicalization diagnostics. The metric called oracle paired-source error compares $T_g\mathcal{G}(a)$ with $T_g u$; it does not evaluate $\mathcal{G}(T_g a)$ and is therefore a diagnostic rather than a new model. Observable canonicalization is also evaluation-only: it reads the total ambient velocity from the input, sets those channels to zero, evaluates the FNO once, and shifts the result by cT. It changes preprocessing but neither retrains the checkpoint nor changes the number of model calls.

6 Experiments

Question. We test whether local orbit consistency reduces symmetry-induced out-of-distribution error and equivariance defect for an unchanged neural-operator backbone. The main experiment uses a periodic 64×64 2D Navier–Stokes vorticity operator with Galilean boosts. The base model is the same FNO2d architecture in every headline row; orbit consistency changes training only and does not add an inference-time module. A near-capacity-matched DeepONet reproduction on the same data tests whether the observed effect requires Fourier layers.

Protocol. We compare four training methods: supervision, supervised Lie augmentation, supervised orbit consistency, and augmentation plus orbit consistency. The headline low-label setting uses 2% labeled data and five seeds. Within both the FNO and DeepONet studies, the forward-evaluation-matched augmentation baseline uses six update steps per epoch, while augmentation plus orbit consistency uses four update steps per epoch with $\lambda_{\text{orb}} = 0.10$. Thus the primary pair uses 12 batch-level operator evaluations per epoch, but not the same optimizer-update or backward-pass count. Every within-backbone row has an unchanged test-time architecture. The FNO headline artifacts are regenerated by `scripts/reproduce_paper_artifacts.py`; reviewer-specific controls, including DeepONet, are regenerated from their recorded runs by `scripts/run_reviewer_followup_experiments.ps1`.

Main N64 Galilean result. Table 5 shows that at 2% labels, augmentation plus orbit consistency improves both paper-facing symmetry metrics over forward-evaluation-matched supervised augmentation. Orbit OOD relative L^2 decreases from 0.5276 ± 0.0525 to 0.3950 ± 0.0334, a 25.1% reduction. Equivariance defect decreases from 0.3816 ± 0.0322 to 0.2677 ± 0.0174, a 29.8% reduction. Relative to the plain supervised

Table 5: N64 Galilean headline at 2% labels over five seeds. All rows use the same FNO2d architecture with 2,668,609 parameters. Lower is better for relative L^2, orbit OOD relative L^2, and equivariance defect.

Method	Steps/epoch	Relative L^2	Orbit OOD L^2	Eq. defect	Latency ms/sample
FNO	4	0.9287 ± 0.0228	0.9568 ± 0.0223	0.6988 ± 0.0173	16.53 ± 2.13
FNO + orbit	4	0.4646 ± 0.0394	0.4952 ± 0.0348	0.3359 ± 0.0170	16.11 ± 1.66
FNO + aug.	6	0.5009 ± 0.0590	0.5276 ± 0.0525	0.3816 ± 0.0322	18.76 ± 7.34
FNO + aug. + orbit	4	$\mathbf{0.3676 \pm 0.0360}$	$\mathbf{0.3950 \pm 0.0334}$	$\mathbf{0.2677 \pm 0.0174}$	16.87 ± 3.19

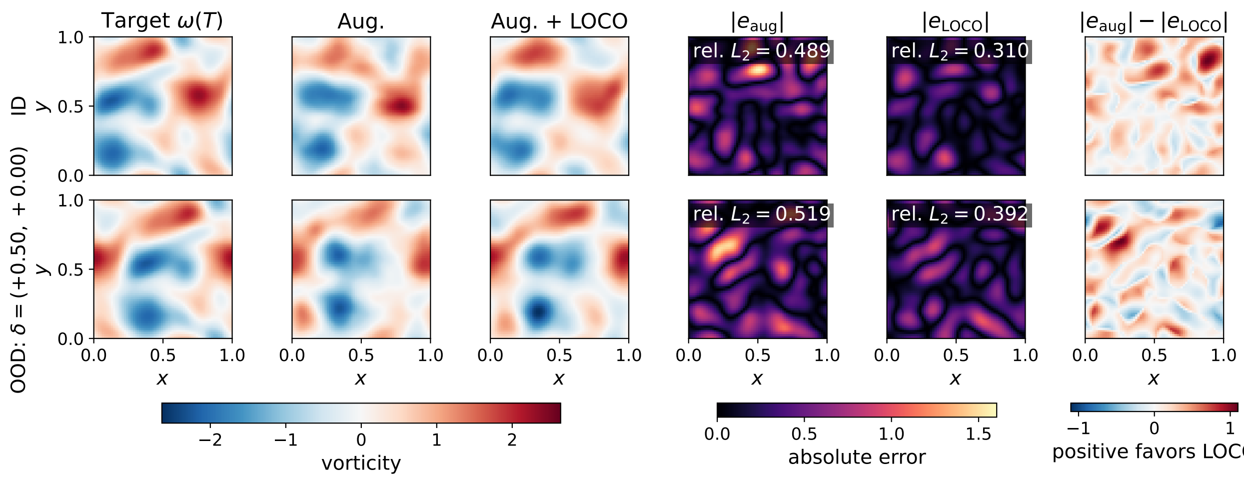

Figure 4: Spatial behavior under a symmetry-induced distribution shift. Rows show the same held-out N64 Navier–Stokes case in distribution (ID) and under a deterministic axis-aligned Galilean stress case $\delta = (0.50, 0)$, twice the componentwise training radius. The ambient frame changes from $(0.073, -0.123)$ to $(0.573, -0.123)$, while the initial vorticity is unchanged. Columns show the target terminal vorticity, direct predictions from forward-evaluation-matched augmentation and augmentation plus LOCO ($\lambda_{\text{orb}} = 0.10$), their pointwise absolute errors, and $|e_{\text{aug}}| - |e_{\text{LOCO}}|$, where positive values favor LOCO. Field and error panels share their respective color ranges across both rows. We use training seed 23 and the test case closest to the median paired fixed-boost improvement over all 256 held-out examples (index 97); the selection rule and checkpoint hashes are recorded with the figure metadata.

FNO baseline, the reductions are 58.7% for orbit OOD error and 61.7% for equivariance defect. Parameter counts are identical across the headline rows.

Qualitative ID/OOD behavior. Figure 4 complements the aggregate metrics with direct field predictions. The large boost increases error for both models, and the residuals trace high-gradient, displaced vorticity interfaces rather than a domain boundary, consistent with the periodic action. In the disclosed representative case, augmentation plus LOCO lowers relative L^2 from 0.489 to 0.310 in distribution and from 0.519 to 0.392 under the fixed boost. Under that boost, LOCO has lower absolute error on 59.7% of pixels, while local negative regions in the gain map show that the improvement is not spatially uniform. Across all 256 test cases for this fixed boost and seed, the mean direct OOD errors are 0.682 for augmentation and 0.556 for augmentation plus LOCO. This axis-aligned case is a qualitative stress test; the independently sampled five-seed severity curve remains the basis of the quantitative claim.

Compute accounting. Table 6 reports training compute for the same headline runs. The augmentation baseline uses 900 optimizer steps and 1800 estimated model forward passes, while augmentation plus orbit consistency uses 600 optimizer steps and the same 1800 estimated model forward passes because each step evaluates the original, augmented, and orbit-transformed inputs. Thus the rows are matched in batch-level forward evaluations and have similar observed wall time; their backward-pass counts differ. Inference latency

Table 6: Training compute accounting for the headline 2% N64 Galilean runs. Forward/backward counts are computed from logged optimizer steps. Transformed-sample counts use the actual alternating minibatch sizes 16 and 4 for the 20-example labeled subset, including iterator restarts; wall-clock minutes are estimated from immutable RNG checkpoint timestamps for these legacy cached runs, and new runs now write `train_wall_seconds` directly. Relative train time is normalized to the supervised augmentation baseline; batch-level forward counts are matched but backward-pass counts differ.

Method	Opt. steps	Model fwds.	Bwd. passes	Aug. labeled samples	Orbit samples	Train min.	Rel. train time	Latency ms/sample
FNO	600	600	600	0	0	14.6 ± 0.4	$0.71\times$	16.53 ± 2.13
FNO + orbit	600	1200	600	0	6000	18.7 ± 0.6	$0.91\times$	16.11 ± 1.66
FNO + aug.	900	1800	900	9000	0	20.6 ± 2.5	$1.00\times$	18.76 ± 7.34
FNO + aug. + orbit	600	1800	600	6000	6000	21.0 ± 2.6	$1.02\times$	16.87 ± 3.19

Table 7: Low-label comparison for forward-evaluation-matched augmentation versus augmentation plus orbit consistency with $\lambda_{\text{orb}} = 0.10$.

Labels	Method	Orbit OOD L^2	Eq. defect	Seeds
1%	Aug.	0.7421 ± 0.0838	0.5080 ± 0.0456	3
1%	Aug. + orbit	$\mathbf{0.6406 \pm 0.0508}$	$\mathbf{0.3999 \pm 0.0197}$	3
2%	Aug.	0.5276 ± 0.0525	0.3816 ± 0.0322	5
2%	Aug. + orbit	$\mathbf{0.3950 \pm 0.0334}$	$\mathbf{0.2677 \pm 0.0174}$	5
5%	Aug.	0.3237 ± 0.0013	0.2427 ± 0.0014	3
5%	Aug. + orbit	$\mathbf{0.2515 \pm 0.0112}$	$\mathbf{0.1621 \pm 0.0078}$	3

remains within run-to-run noise because the deployed map is still the same FNO2d backbone.

Low-label behavior. Figure 5 and Table 7 show lower mean errors at each tested low-label fraction. Against forward-evaluation-matched augmentation, $\lambda_{\text{orb}} = 0.10$ reduces orbit OOD error by 13.7%, 25.1%, and 22.3% at 1%, 2%, and 5% labels respectively. The corresponding equivariance-defect reductions are 21.3%, 29.8%, and 33.2%. These finite settings support a low-label benefit but do not establish a general sample-complexity law.

Backbone portability and dependence. Table 8 reproduces the full four-method experiment with a fixed-grid branch–trunk DeepONet whose parameter count differs from FNO2d by 0.73%. Relative to forward-evaluation-matched DeepONet augmentation, adding normalized LOCO lowers ID error from 0.9553 ± 0.0335 to 0.8862 ± 0.0362 (7.2%), orbit OOD error from 0.9652 ± 0.0308 to 0.8972 ± 0.0340 (7.0%), and equivariance defect from 0.6988 ± 0.0219 to 0.4921 ± 0.0295 (29.6%). The matched-seed LOCO-minus-augmentation differences are -0.06911 for ID error (95% CI $[-0.08038, -0.05784]$), -0.06799 for orbit OOD error ($[-0.08087, -0.05510]$), and -0.20674 for defect ($[-0.25557, -0.15791]$). All five seed-level differences have the favorable sign for all three metrics. Thus the benefit transfers to the tested second neural-operator family without changing its inference graph. The predictive reductions are nevertheless smaller than the corresponding FNO reductions (7.2% versus 26.6% for ID and 7.0% versus 25.1% for OOD), while the relative defect reductions are similar (29.6% versus 29.8%). These are descriptive within-task comparisons, not a cross-backbone significance test, and the tested DeepONet is fixed-grid and less accurate overall.

Table 9 gives two secondary reference points. In a five-seed, forward-evaluation-matched rMD17 check, adding normalized LOCO to a fixed-molecule MLP reduces force MAE by 2.7%, transformed-input force MAE by 2.9%, and force equivariance defect by 64.6%; the paired 95% intervals for all three deltas exclude zero. This is a cross-domain mechanism check, not a neural operator or an OOD-severity result. Conversely, a supervised D4 G-FNO already has numerical defect below 10^{-7}, leaving essentially no same-group consistency signal; its predictive error is higher in the reported, non-capacity-matched configuration. We therefore claim architecture-independent applicability of the objective, not architecture-independent effect size or uniform gains across neural-operator families.

Table 8: DeepONet reproduction on the 2%-label N64 Galilean task over the same training seeds as the FNO comparison, $\{23, 31, 47, 59, 71\}$. All rows use the same 2,688,033-parameter architecture. The augmentation and augmentation-plus-normalized-LOCO rows are matched at 12 batch-level model evaluations per epoch; optimizer and backward-pass counts differ and are shown. Entries are mean ± sample standard deviation. Latency and wall time remain available in the per-run manifest; seed shards ran concurrently, so they are not used as efficiency comparisons.

Method	Steps/epoch	Model fwds., total	Bwd. passes, total	ID L^2	Orbit OOD L^2	Eq. defect
DeepONet	4	600	600	0.9949 ± 0.0081	1.0013 ± 0.0072	0.5821 ± 0.0429
DeepONet + normalized LOCO	4	1200	600	0.9150 ± 0.0318	0.9236 ± 0.0300	0.5545 ± 0.0229
DeepONet + aug.	6	1800	900	0.9553 ± 0.0335	0.9652 ± 0.0308	0.6988 ± 0.0219
DeepONet + aug. + normalized LOCO	4	1800	600	$\mathbf{0.8862 \pm 0.0362}$	$\mathbf{0.8972 \pm 0.0340}$	$\mathbf{0.4921 \pm 0.0295}$

Table 9: Backbone scope over five seeds. Panel A directly applies LOCO to a non-Fourier, non-equivariant fixed-molecule MLP; ID and transformed errors are force MAE, and both rows use 48 batch-level model evaluations per epoch. The LOCO row uses the disclosed composite rigid-motion step normalization. Panel B uses relative L^2 error and contrasts an FNO training loss with a supervised hard-equivariant D4 G-FNO. The G-FNO is a saturation reference, not a LOCO-trained row, and is not parameter- or mode-matched. Metrics must not be compared across panels.

Task / backbone	Training	ID error	Transformed error	Eq. defect	Parameters	Seeds
Panel A: rMD17 ethanol force prediction, 500 labels						
MLP	augmentation	19.9135 ± 0.0895	20.0535 ± 0.0812	0.7372 ± 0.0139	57,755	5
MLP	augmentation + normalized LOCO	$\mathbf{19.3686 \pm 0.1389}$	$\mathbf{19.4698 \pm 0.1342}$	$\mathbf{0.2608 \pm 0.0044}$	57,755	5
Panel B: unboosted N64 Navier–Stokes under D4						
FNO2d	augmentation	0.06542 ± 0.00369	0.06538 ± 0.00366	0.04570 ± 0.00397	2,668,513	5
FNO2d	augmentation + LOCO	0.06462 ± 0.00331	0.06459 ± 0.00330	0.04499 ± 0.00383	2,668,513	5
D4 G-FNO	supervised	0.09283 ± 0.00537	0.09283 ± 0.00537	$(7.88 \pm 0.32) \times 10^{-8}$	350,280	5

Solver closure and supporting Burgers evidence. Explicit re-solves of 32 transformed test examples verify the coded Galilean identity to numerical precision: the maximum relative closure residual is 6.22×10^{-7} for N64 Navier–Stokes through component radius 0.50 and 7.51×10^{-7} for N128 Burgers through boost radius 1.05. On the learned Burgers operator, Table 10 shows that augmentation already nearly saturates the symmetry. Adding LOCO changes mean orbit-shift error from 0.00405 to 0.00381, but the paired interval crosses zero; the defect changes from 0.00265 to 0.00228, with its paired interval below zero, while mean ID error worsens from 0.00251 to 0.00265. We therefore use Burgers as an exact-action supporting check, not a second headline predictive result or a uniform accuracy improvement.

Finite versus tangent and raw consistency. Table 11 compares the finite LOCO term with an explicit JVP tangent penalty and a plain finite-transform consistency loss. On five paired seeds, normalized finite LOCO minus augmentation plus tangent is -0.04734 in ID error (95% CI $[-0.05494, -0.03974]$), -0.05397 in orbit-shift error ($[-0.06037, -0.04758]$), and -0.04799 in equivariance defect ($[-0.05209, -0.04390]$). The tested finite objective therefore achieves lower means, consistent with useful information away from the identity. It does not causally isolate higher-order orbit information, however: the explicit JVP has a different computational profile and only one tangent weight was tested. The comparison therefore does not establish universal superiority over tangent propagation. On the three seeds shared with the approximately scale-matched raw finite control, normalization changes ID error by -0.01232 (95% CI $[-0.02997, 0.00533]$), orbit-shift error by -0.00990 ($[-0.02448, 0.00467]$), and defect by -0.00016 ($[-0.01534, 0.01503]$). All three intervals cross zero. Thus this control does not resolve a difference between normalized and raw finite penalties at this radius and approximate scale match; the intervals are not an equivalence or noninferiority test. It therefore does not by itself show that normalization improves predictive performance. The normalization's specific contribution is to remove the leading quadratic coupling between objective scale and sampled step radius.

OOD severity. Figure 7 and Table 16 evaluate Galilean boosts from below the training radius to twice the training radius. The augmentation-plus-orbit model maintains a lower orbit OOD error at every tested

Table 10: Supporting 1D Burgers exact-symmetry check over three matched seeds. The orbit-shift error difference has a paired interval crossing zero, while the equivariance-defect reduction does not.

Training	ID L^2	Orbit-shift L^2	Eq. defect
Augmentation	0.002512 ± 0.000146	0.004049 ± 0.000326	0.002653 ± 0.000326
Augmentation + LOCO	0.002655 ± 0.000003	0.003807 ± 0.000153	0.002283 ± 0.000215

Table 11: Finite-objective comparison on the N64 Galilean task at 2% labels. Entries are mean ± sample standard deviation within each panel. Panel A uses five matched seeds; Panel B repeats normalized finite LOCO on the three seeds available for the raw finite control. The augmentation baseline and both finite objectives use 12 batch-level model-forward evaluations per epoch. The tangent objective uses an explicit JVP and is not compute-matched. The raw finite weight 2.4 is an approximate leading-order average-scale match to normalized finite weight 0.10, not a tuned equivalence.

Method	Weight	Step normalization	JVP	ID L^2	Orbit-shift L^2	Eq. defect	Seeds
Panel A: finite versus tangent, five matched seeds							
Augmentation	$\lambda_{\text{aug}} = 1$	n/a	no	0.5009 ± 0.0590	0.5276 ± 0.0525	0.3816 ± 0.0322	5
Augmentation + tangent	$\lambda_{\text{tangent}} = 0.1$	n/a	yes	0.4150 ± 0.0363	0.4490 ± 0.0336	0.3157 ± 0.0203	5
Augmentation + normalized finite	$\lambda_{\text{orb}} = 0.1$	$\epsilon^2 + \eta$	no	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	5
Panel B: normalized versus raw finite, three matched seeds							
Augmentation + normalized finite	$\lambda_{\text{orb}} = 0.1$	$\epsilon^2 + \eta$	no	0.3674 ± 0.0136	0.3915 ± 0.0151	0.2630 ± 0.0087	3
Augmentation + raw finite	$\lambda_{\text{raw}} = 2.4$	none	no	0.3797 ± 0.0206	0.4014 ± 0.0210	0.2632 ± 0.0148	3

boost. The relative OOD-error reductions are 26.4%, 25.8%, 25.3%, 23.6%, and 20.3% at maximum boosts 0.10, 0.20, 0.25, 0.35, and 0.50. Equivariance-defect reductions follow the same trend, from 33.6% at boost 0.10 to 23.6% at boost 0.50.

Canonicalization calibration. Table 13 reports a privileged relative-boost calibration, not a deployable analytic canonicalization baseline. In the OOD severity protocol, each test input a is paired with a sampled relative boost δ, producing $T_\delta a$. The calibration uses this sampled δ and the paired in-distribution prediction $\mathcal{G}_\theta(a)$, then reports $T_\delta \mathcal{G}_\theta(a)$. This is useful for measuring how much of the direct OOD error comes from motion along the tested orbit, but it assumes access to the paired source input and sampled relative transform. It should therefore not be confused with the observable-frame analytic baseline, which reads the absolute ambient velocity c from the input channels and canonicalizes a single test input. At maximum boost 0.50, augmentation plus orbit consistency has direct OOD error 0.4910 ± 0.0216, improving substantially over augmentation 0.6158 ± 0.0384 while remaining above its privileged calibration value 0.3676 ± 0.0360.

Association between defect and OOD error. Figure 6 and Table 14 report a diagnostic association rather than a causal or out-of-sample prediction result. Across 90 cached N64 Galilean observations, defect correlates strongly with orbit OOD relative L^2 (Pearson $r = 0.851$, Spearman $\rho = 0.838$); after residualizing source, method, label fraction, maximum boost, and λ_{orb}, the partial Pearson correlation remains $r = 0.779$. The association also appears within the OOD severity curve alone ($r = 0.761$). Because all observations share datasets and pipelines, these results show that orbit consistency and OOD behavior track one another in this benchmark, not that defect is a generally validated predictor.

Boundary-condition diagnostic. Table 12 highlights a limitation rather than extending the exact-symmetry claim. Fixed Dirichlet walls break global translation equivariance, so all checkpoints are re-evaluated using the same shift draws and the same full-domain or interior support. Masked Aug.+LOCO gives the lowest valid-overlap pseudo-target error and defect, but its full-domain pseudo-target score is worse than unmasked Aug.+LOCO, and both regularized variants have worse genuine ID error than the baseline. The mask therefore localizes where an approximate constraint is useful; it does not turn the fixed-wall problem into a translation symmetry. This diagnostic motivates checking closure of the full equation, geometry, boundary data, and observation map before applying LOCO.

Table 12: Fixed-boundary heat-equation diagnostic over three seeds. Every checkpoint is re-evaluated using identical transform draws and the same scoring support. "Approximate-action pseudo-target" compares with a shifted cached target, not a newly solved transformed problem. The overlap columns use the same transform-dependent source-overlap mask for every method, with source margin 0.05. Lower is better.

Training	ID L^2	Approx.-action pseudo-target, full	Defect, full	Approx.-action pseudo-target, overlap	Defect, overlap
Baseline	0.0270 ± 0.0012	0.0451 ± 0.0007	0.0341 ± 0.0008	0.0332 ± 0.0015	0.0243 ± 0.0006
Aug.+LOCO, unmasked	0.0347 ± 0.0019	0.0433 ± 0.0014	0.0246 ± 0.0008	0.0336 ± 0.0016	0.0149 ± 0.0006
Aug.+LOCO, masked	0.0363 ± 0.0012	0.0447 ± 0.0008	0.0301 ± 0.0006	$\mathbf{0.0230 \pm 0.0014}$	$\mathbf{0.0117 \pm 0.0004}$

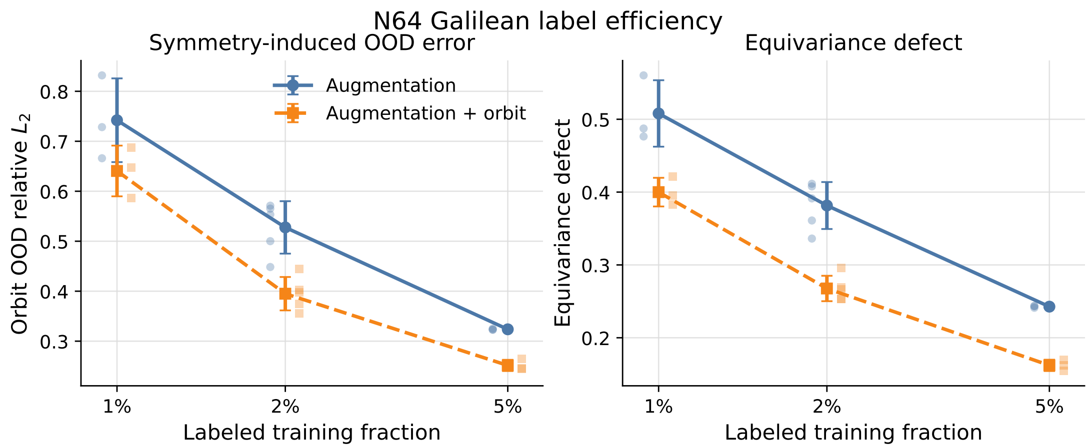

Figure 5: N64 Galilean label efficiency. Error bars show standard deviation across seeds and faint points show individual seeds. Augmentation plus orbit consistency improves orbit OOD error and equivariance defect at 1%, 2%, and 5% labels.

Ablations and scope. Table 15 reports the orbit-weight ablation at 2% labels. We keep $\lambda_{\text{orb}} = 0.10$ as the broader-experiment setting, while the completed high-weight sweep shows $\lambda_{\text{orb}} = 0.30$ as the strongest tested value. Semi-supervised joint orbit training is useful as an appendix result because it reduces equivariance defect, but it is not the strongest predictive-error result. We do not include a 3D PDE in the main study: a same-resolution 3D analogue would multiply raw field size by 64 and introduce a new solver and validation burden. The current contribution is deliberately narrow: exact-symmetry validation plus a reproducible 2D Galilean low-label result.

7 Discussion and Limitations

The evidence supports a narrow conclusion: on the solver-validated periodic N64 Galilean task, LOCO improves the tested FNO's symmetry-shifted error and equivariance defect, with the largest observed gains at low label counts. Burgers and D4 are supporting checks rather than equally strong replications. The five-seed, near-capacity-matched DeepONet reproduction gives the same direction for ID error, orbit OOD error, and defect on the same dataset. Its predictive reductions are 7.2% and 7.0%, compared with 26.6% and 25.1% for FNO, while its 29.6% defect reduction is close to FNO's 29.8%. This is one fixed-grid, non-FNO branch–trunk operator check, not evidence across operator families and equations. The normalized molecular MLP experiment is a secondary cross-domain mechanism check, but it is neither a neural operator nor a competitive atomistic architecture. Conversely, the D4 G-FNO illustrates backbone dependence: a hard equivariant architecture already saturates the same-group defect and leaves no meaningful consistency signal. Further non-Fourier operator families, resolutions, and datasets remain future work.

The tangent and raw finite-transform controls isolate only the tested coefficients and schedules. They distinguish explicit identity-tangent regularization from finite group evaluations, and test the leading-order role of step normalization, but they do not exhaustively tune every objective. Likewise, matching batch-level

Table 13: Privileged relative-boost calibration at 2% labels. Direct OOD evaluates the model on the boosted input. The privileged calibration uses the sampled relative boost from the evaluation protocol and the paired in-distribution prediction, so it is not an observable-frame analytic baseline. Lower is better.

Max boost	Aug. direct OOD	Aug. priv. calib.	Aug. + orbit direct OOD	Aug. + orbit priv. calib.
0.10	0.5050 ± 0.0574	0.5009 ± 0.0590	$\mathbf{0.3715 \pm 0.0349}$	0.3676 ± 0.0360
0.20	0.5176 ± 0.0543	0.5009 ± 0.0590	$\mathbf{0.3840 \pm 0.0331}$	0.3676 ± 0.0360
0.25	0.5277 ± 0.0523	0.5009 ± 0.0590	$\mathbf{0.3943 \pm 0.0318}$	0.3676 ± 0.0360
0.35	0.5555 ± 0.0476	0.5009 ± 0.0590	$\mathbf{0.4242 \pm 0.0285}$	0.3676 ± 0.0360
0.50	0.6158 ± 0.0384	0.5009 ± 0.0590	$\mathbf{0.4910 \pm 0.0216}$	0.3676 ± 0.0360

Table 14: Correlation between equivariance defect and orbit OOD relative L^2 across cached N64 Galilean seeds, label fractions, orbit weights, and boost severities. Partial Pearson r residualizes source, method, label fraction, maximum boost, and λ_{orb}.

Scope	Obs.	Pearson r	Spearman ρ	Partial r
All	90	0.851	0.838	0.779
Train/lambda/label	40	0.990	0.990	0.976
OOD severity	50	0.761	0.754	0.516

forward evaluations does not match backward passes or all transformation overhead. LOCO adds training work; its deployment claim is only that it adds no inference-time module, parameter, or additional operator call.

Finally, the method is only as sound as the action supplied to it. Exact use requires closure of the equation, geometry, coefficients, forcing and boundary data, state representation, and observation map under the transformation. The Dirichlet heat diagnostic shows that a spatial mask can localize an approximate constraint but cannot restore a broken global symmetry. Applying LOCO to a plausible but invalid action may degrade predictive accuracy, so solver-level identity checks and boundary diagnostics should precede long training runs.

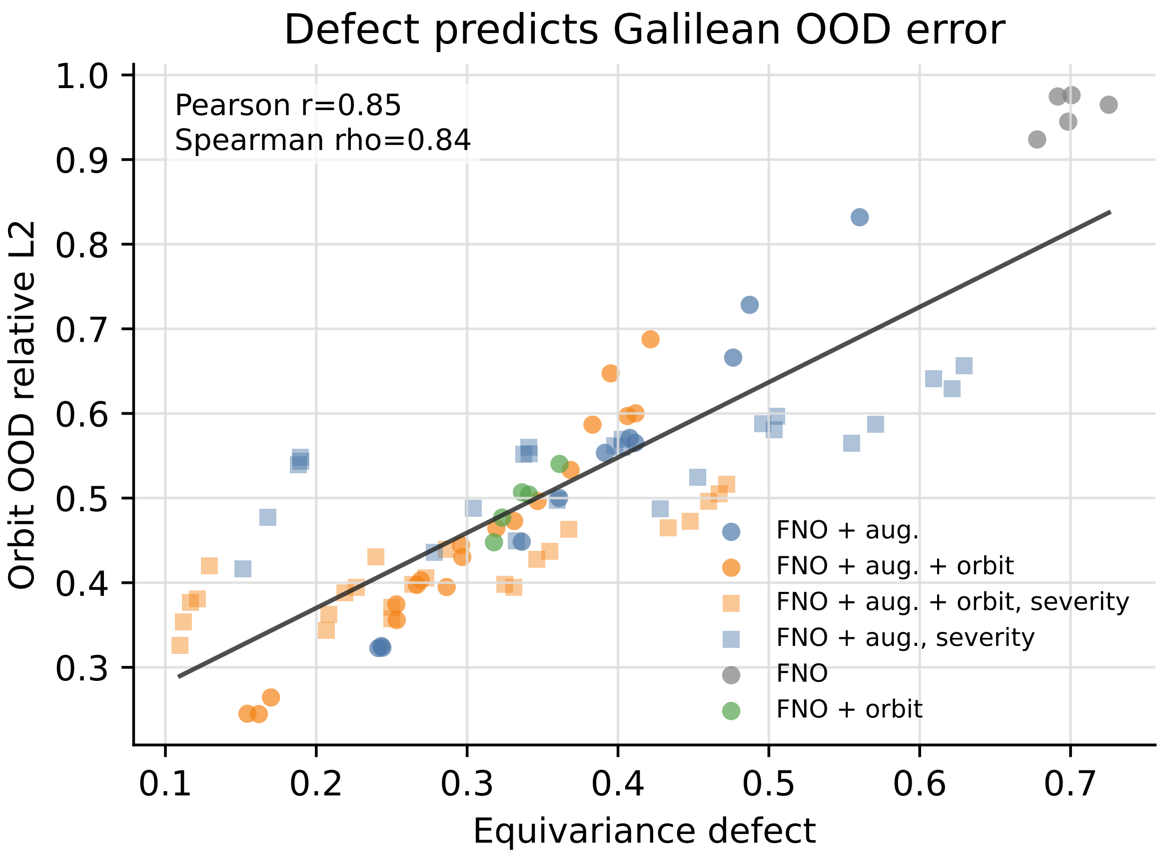

Figure 6: Association between equivariance defect and Galilean orbit OOD error across cached N64 seeds, label fractions, orbit weights, and boost severities. Circles are trained-model evaluations and squares are severity evaluations. Shared datasets and pipelines make this a diagnostic association rather than a causal or out-of-sample prediction result.

Table 15: Orbit-weight ablation at 2% labels. $\lambda_{\mathrm{orb}} = 0.10$ is used for the broader experiments; $\lambda_{\mathrm{orb}} = 0.30$ is the strongest among the tested values.

λ_{orb}	ID L^2	Orbit OOD L^2	Eq. defect	Seeds
0.10	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	5
0.20	0.3291 ± 0.0318	0.3564 ± 0.0304	0.2268 ± 0.0165	5
0.30	$\mathbf{0.3149 \pm 0.0298}$	$\mathbf{0.3416 \pm 0.0289}$	$\mathbf{0.2044 \pm 0.0160}$	5

Table 16: Relative reductions of augmentation plus orbit consistency over forward-evaluation-matched augmentation across Galilean OOD severities at 2% labels.

Max boost	OOD reduction	Defect reduction
0.10	26.4%	33.6%
0.20	25.8%	31.3%
0.25	25.3%	30.2%
0.35	23.6%	27.8%
0.50	20.3%	23.6%

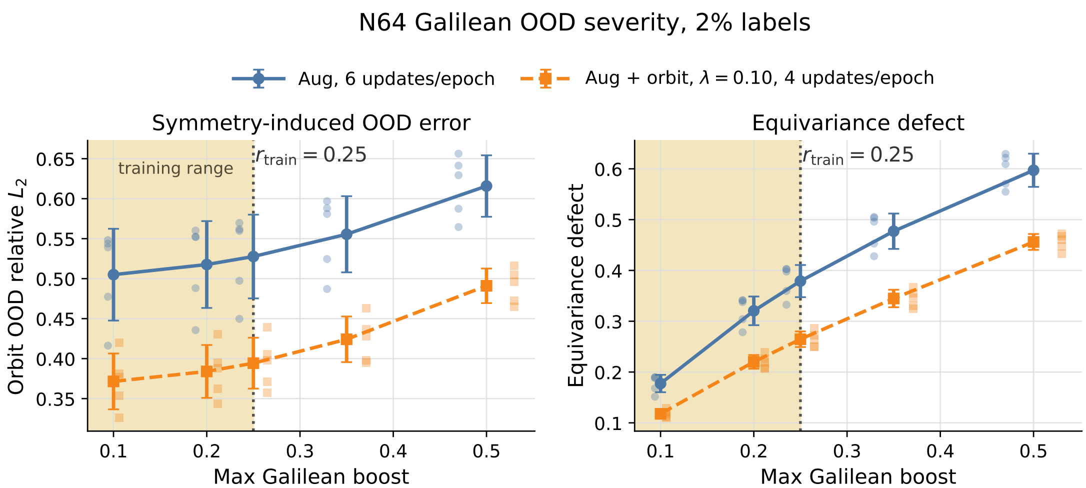

Figure 7: OOD severity at 2% labels. The dashed line marks the componentwise training radius. Augmentation plus orbit consistency remains better than forward-evaluation-matched augmentation from below the training radius through twice the training radius.

References

[1] Tara Akhound-Sadegh, Laurence Perreault-Levasseur, Johannes Brandstetter, Max Welling, and Siamak Ravanbakhsh. Lie point symmetry and physics-informed networks. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8493c860bec41705f7743d5764301b94-Abstract-Conference.html.

[2] Simon Batzner, Albert Musaelian, Lixin Sun, Mario Geiger, Jonathan P. Mailoa, Mordechai Kornbluth, Nicola Molinari, Tess E. Smidt, and Boris Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13:2453, 2022. doi: 10.1038/s41467-022-29939-5. URL https://doi.org/10.1038/s41467-022-29939-5.

[3] Boris Bonev, Thorsten Kurth, Christian Hundt, Jaideep Pathak, Maximilian Baust, Karthik Kashinath, and Anima Anandkumar. Spherical Fourier neural operators: Learning stable dynamics on the sphere. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2806–2823. PMLR, 2023. URL https://proceedings.mlr.press/v202/bonev23a.html.

[4] Johannes Brandstetter, Max Welling, and Daniel E Worrall. Lie point symmetry data augmentation for neural PDE solvers. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 2241–2256. PMLR, 17–23 Jul 2022. URL https://proceedings.mlr.press/v162/brandstetter22a.html.

[5] Gengxiang Chen, Xu Liu, Qinglu Meng, Lu Chen, Changqing Liu, and Yingguang Li. Learning neural operators on riemannian manifolds. *National Science Open*, 3:20240001, 2024. doi: 10.1360/nso/20240001. URL https://doi.org/10.1360/nso/20240001.

[6] Anders S. Christensen and O. Anatole von Lilienfeld. On the role of gradients for machine learning of molecular energies and forces. *Machine Learning: Science and Technology*, 1(4):045018, 2020. doi: 10.1088/2632-2153/abba6f. URL https://doi.org/10.1088/2632-2153/abba6f.

[7] Taco Cohen and Max Welling. Group equivariant convolutional networks. In Maria Florina Balcan and Kilian Q. Weinberger, editors, *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2990–2999. PMLR, 2016. URL https://proceedings.mlr.press/v48/cohenc16.html.

[8] Taco Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral CNN. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2019. URL https://proceedings.mlr.press/v97/cohen19d.html.

[9] Vignesh Gopakumar, Stanislas Pamela, Lorenzo Zanisi, Zongyi Li, Ander Gray, Daniel Brennand, Nitesh Bhatia, Gregory Stathopoulos, Matt J. Kusner, Marc Peter Deisenroth, Anima Anandkumar, JOREK Team, and MAST Team. Plasma surrogate modelling using fourier neural operators. *Nuclear Fusion*, 64(5):056025, 2024. doi: 10.1088/1741-4326/ad313a. URL https://doi.org/10.1088/1741-4326/ad313a.

[10] Jiequn Han, Xu-Hui Zhou, and Heng Xiao. An equivariant neural operator for developing nonlocal tensorial constitutive models. *Journal of Computational Physics*, 488:112243, 2023. doi: 10.1016/j.jcp.2023.112243. URL https://doi.org/10.1016/j.jcp.2023.112243.

[11] Jacob Helwig, Xuan Zhang, Cong Fu, Jerry Kurtin, Stephan Wojtowytsch, and Shuiwang Ji. Group equivariant Fourier neural operators for partial differential equations. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 12907–12930. PMLR, 23–29 Jul 2023. URL https://proceedings.mlr.press/v202/helwig23a.html.

[12] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to pdes. *Journal of Machine Learning Research*, 24(89):1–97, 2023. URL https://www.jmlr.org/papers/v24/21-1524.html.

[13] Samuel Lanthaler, Roberto Molinaro, Patrik Hadorn, and Siddhartha Mishra. Nonlinear reconstruction for operator learning of pdes with discontinuities. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=CrfhZAsJDsZ.

[14] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=c8P9NQVtmnO.

[15] Zongyi Li, Daniel Zhengyu Huang, Burigede Liu, and Anima Anandkumar. Fourier neural operator with learned deformations for pdes on general geometries. *Journal of Machine Learning Research*, 24 (388):1–26, 2023. URL https://www.jmlr.org/papers/v24/23-0064.html.

[16] Zongyi Li, Nikola Borislavov Kovachki, Chris Choy, Boyi Li, Jean Kossaifi, Shourya Otta, Mohammad Amin Nabian, Maximilian Stadler, Christian Hundt, Kamyar Azizzadenesheli, and Anima Anandkumar. Geometry-informed neural operator for large-scale 3d pdes. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/70518ea42831f02afc3a2828993935ad-Abstract-Conference.html.

[17] Zongyi Li, Hongkai Zheng, Nikola Borislavov Kovachki, David Jin, Haoxuan Chen, Burigede Liu, Andrew Stuart, Kamyar Azizzadenesheli, and Anima Anandkumar. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024. doi: 10.1145/3648506. URL https://doi.org/10.1145/3648506.

[18] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021. doi: 10.1038/s42256-021-00302-5. URL https://doi.org/10.1038/s42256-021-00302-5.

[19] Ruben Ohana, Michael McCabe, Lucas Thibaut Meyer, Rudy Morel, Fruzsina Julia Agocs, Miguel Beneitez, Marsha Berger, Blakesley Burkhart, Stuart B. Dalziel, Drummond Buschman Fielding, Daniel Fortunato, Jared A. Goldberg, Keiya Hirashima, Yan-Fei Jiang, Rich Kerswell, Suryanarayana Maddu, Jonah M. Miller, Payel Mukhopadhyay, Stefan S. Nixon, Jeff Shen, et al. The well: A large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. URL https://openreview.net/forum?id=00Sx577BT3.

[20] Zakhar Shumaylov, Peter Zaika, James Rowbottom, Ferdia Sherry, Melanie Weber, and Carola-Bibiane Schönlieb. Lie algebra canonicalization: Equivariant neural operators under arbitrary lie groups. In *International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=7PLpiVdnUC.

[21] Patrice Simard, Bernard Victorri, Yann LeCun, and John Denker. Tangent prop: A formalism for specifying selected invariances in an adaptive network. In *Advances in Neural Information Processing Systems*, volume 4, 1991. URL https://proceedings.neurips.cc/paper/1991/hash/65658fde58ab3c2b6e5132a39fae7cb9-Abstract.html.

[22] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/0a9747136d411fb83f0cf81820d44afb-Abstract-Datasets_and_Benchmarks.html.

[23] Amy Xiang Wang, Zakhar Shumaylov, Peter Zaika, Ferdia Sherry, and Carola-Bibiane Schönlieb. Generalized lie symmetries in physics-informed neural operators, 2025. URL https://arxiv.org/abs/2502.00373.

[24] Maurice Weiler and Gabriele Cesa. General $E(2)$-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1911.08251.

[25] Jiaxiao Xu, Changhong Mou, Yeyu Zhang, and Fengxiang He. Physics-aligned canonical equivariant fourier neural operator under symmetry-induced shifts, 2026. URL https://arxiv.org/abs/2605.18606.